Dual-UNet++ Framework for Robust Single-Frame Echocardiographic Dehazing

Dominik Fromme[1[0009−0003−8312−1229]], Matti Kaupenjohann[1[0000−0001−9663−9631]], Tim Streckert[1[0009−0002−5970−4101]], and Jörg Thiem[1[0000−0002−8586−2747]]

University of Applied Sciences and Arts Dortmund, Germany
{dominik.fromme,matti.kaupenjohan,tim.streckert,joerg.thiem}@fh-dortmund.de

Abstract. The Dehazing Echocardiography Challenge 2025 required single-frame denoising, disallowing temporal information. Nevertheless, we exploited sequence structure during training by applying log-PCA decomposition and using the mid-order reconstruction (up to $k = 35$ components) as supervised target. This strategy enabled the network to predict temporally consistent reconstructions from individual frames, transferring temporal coherence into a single-frame setting. High-order components served as structured noise sources, allowing generation of diverse synthetic corruptions and improving robustness.
To enforce anatomical separation, we added an auxiliary segmentation branch with stochastic threshold sampling of PCA reconstructions, which acted as target augmentation resulting in a stable structure-background discrimination. Sparse ROI labels were densified via interpolation and further refined using a dedicated auxiliary network.
Our dual UNet++ framework with ResNet backbones achieved a final challenge score of 84.28, with competitive performance across FID, CNR, gCNR, and Dice. These results demonstrate that temporally informed PCA supervision, structured noise augmentation, and stochastic segmentation targets are effective strategies for robust single-frame echocardiographic dehazing.

Keywords: Dehazing Echocardiography Challenge 2025 · Ultrasound Denoising · Medical Image Enhancement · Principal Component Analysis (PCA)· Structured Noise Augmentation · UNet++ · Stochastic Label Thresholding

1 Introduction

Echocardiography is the most widely used imaging modality for assessing cardiac structure and function due to its real-time acquisition, portability, and safety. However, image quality is often degraded by noise, artifacts, and patient-specific factors, which hampers both clinical interpretation and automated analysis.

The Dehazing Echocardiography Challenge 2025 is designed to address this problem by evaluating methods for improving the visibility of cardiac structures in noisy ultrasound data [14, 4, 7, 17, 13, 5].

In this work, we propose a PCA-based [8] denoising framework [11] that leverages temporal information during training while producing single-frame predictions. Specifically, log-PCA decomposition of video sequences defines temporally consistent reconstruction targets and extracts structured noise components for data augmentation. In addition, ROI interpolation and auxiliary segmentation increase label density and robustness. The final framework consists of two UNet++ models [18, 19]: the first predicts PCA reconstructions and segmentation masks, while the second refines these outputs through residual correction to better match the challenge objectives.

2 Methods

2.1 Data and Preprocessing

Task and Data The challenge provides short, patient-specific echocardiography sequences (typically 60 frames per case). Although evaluation requires single-frame denoising and the organizers randomize frame order across patients at inference, we exploit temporal ordering during training to densify labels. Two anatomical classes are considered: A (ventricular septum) and B (ventricle). Sparse ROI masks are available on every 10th frame (key frames) of noisy images.

ROI densification via distance-map interpolation To obtain dense annotations, we interpolate between consecutive key frames. Signed distance maps are generated for each ROI and linearly interpolated over time. The binary masks are then recovered by thresholding. This yields per-frame ROIs for both classes across all sequences. As these labels originate from interpolation rather than manual annotation, we treat them as weak in subsequent training.

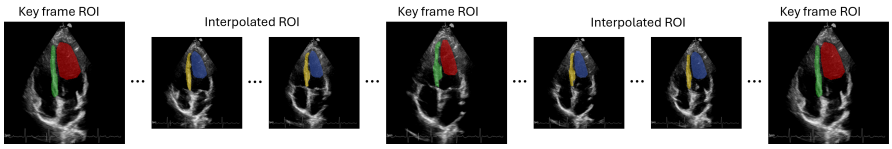

Fig. 1. Showing keyframe images with GT-ROI (green, red) and the interpolated ROIs between keyframes (yellow, blue).

ROI extension via auxiliary network In addition, we train an auxiliary segmentation network (H) to generate ROIs for clean images, enabling their integration into later training stages. This step improves label coverage and consistency across noisy and clean data.

2.2 PCA-Based Representation

Prior to decomposition, each frame I_t is logarithmically transformed to reduce multiplicative speckle noise and stabilize variance:

$$I_t^{\log} = \log(1 + I_t). \tag{1}$$

Let $\{I_t^{\log}\}_{t=1}^{T}$ be the log-transformed sequence of a patient and let $\mu \in \mathbb{R}^{H \times W}$ denote the per-patient mean image in log space. Perform PCA on the mean-centered data to obtain principle components $\{v_j\}_{j=1}^{J}$ and per-frame scores $\{s_{t,j}\}$. The mid-order reconstruction used as the supervised target is then defined as:

$$I_t^{pca} = \mu + \sum_{j=1}^{k} s_{t,j}\, v_j, \qquad k = 35. \tag{2}$$

We empirically select $k = 35$ as a trade-off where low-order components capture static content, mid-order components retain physiologically meaningful motion and higher-order components are predominantly noise.

To simulate realistic corruptions, we additionally exploit the high-order PCA reconstruction components ($k \approx 45{-}60$) that mainly represent stochastic variation. The synthetic noise map is constructed as

$$N_t = \sum_{j \in I} \epsilon_j\, m_j\, \tilde{v}_j, \qquad \epsilon_j \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

where $\tilde{v}_j$ is the normalized PCA basis image of component j, m_j a magnitude factor, and σ the stochastic variance. A new augmented noise input image is then obtained as

$$I_t^{aug} = \mathrm{clip}_{[0,255]}\big(I_t + \gamma\, N_t\big), \tag{4}$$

with γ as a global scaling factor controlling the overall noise level. In practice, I_t and N_t are normalized to $[0, 1]$ before augmentation, and the final result is rescaled to $[0, 255]$ for consistency with the challenge format.

This formulation enables single-frame predictions to approximate temporally consistent PCA reconstructions, while high-order components serve as structured noise sources that expand the training distribution and improve robustness against patient-specific noise.

Notice: To emphasize septum–ventricle discrimination, PCA reconstructions are masked by setting values inside the ventricle ROI to zero whenever a mask is available.

2.3 Model Architecture

Our framework consists of two UNet++ models with ResNet backbones [3], initialized with ImageNet-pretrained weights [1] and decoder channels set to [256,128,64,32,16], designed to jointly exploit PCA-based supervision and segmentation cues.

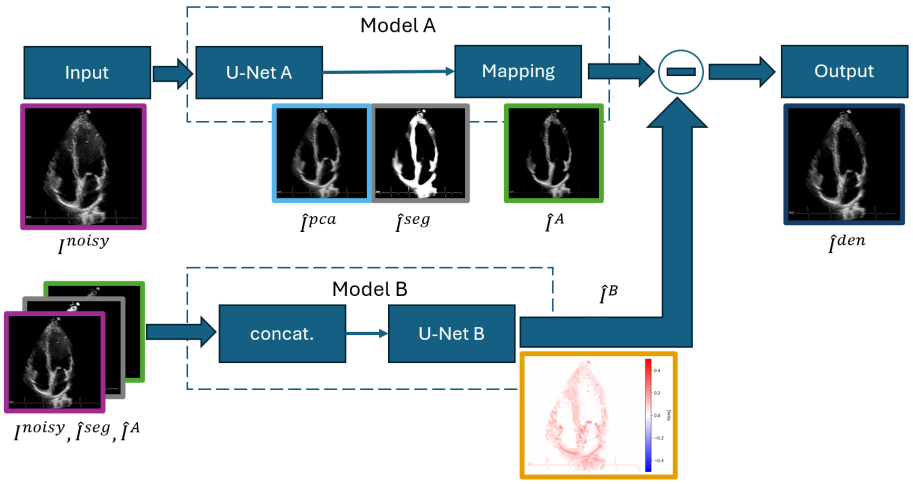

Fig. 2. Overall Model

Model A Model A is a UNet++ with a ResNet34 encoder (26.3 M parameters) that takes a noisy input image I^{noisy} and predicts two outputs: (i) a regression image $\hat{I}^{\text{pca}}$ approximating the PCA reconstruction and (ii) a binary segmentation of the cardiac structures $\hat{I}^{\text{seg}}$.

Binary target masks are dynamically generated during training by thresholding the PCA reconstruction $\hat{I}^{\text{pca}}$ (up to first $k = 35$ components). Since the images are highly corrupted by noise and even human observers can only define broad threshold ranges, we replace fixed rules thresholds (e.g. Otsu) with *random threshold sampling* within these bounds, which also acts as target augmentation, improves generalization and enforcing robust structure–background discrimination.

The final denoised output of Model A, denoted $\hat{I}^{\text{A}}$, is obtained by applying a differentiable pixel-wise nonlinear remapping to the segmentation output and multiplying it element-wise with the predicted PCA reconstruction. Specifically, the nonlinear mapping $\mathcal{M}(\cdot)$ is defined as:

$$\mathcal{M}(x) = m_{\min} + (m_{\max} - m_{\min})\, \sigma\big(k(x - 0.5)\big), \quad x \in [0, 1] \tag{5}$$

where $\sigma(\cdot)$ denotes the sigmoid function, k is the slope parameter, and $m_{\min}$,$m_{\max}$ are the lower and upper remapping bounds. In our experiments, we used $k = 2.0$, $m_{\min} = 0.2$, and $m_{\max} = 1.01$. The resulting denoised image is then computed as:

$$\hat{I}^{\text{A}} = \hat{I}^{\text{pca}} \odot \mathcal{M}\Big(\hat{I}^{\text{seg}}\Big), \tag{6}$$

Model B Model B is a UNet++ with a ResNet18 encoder (16.2 M parameters) that takes as input the concatination of the noisy input image I^{noisy}, the binary

segmentation $\hat{I}^{\text{seg}}$ and model A output $\hat{I}^{\text{A}}$. It returns a delta image $\hat{I}^{\text{B}}$ that is subtracted from $\hat{I}^{\text{A}}$ to return the final output $\hat{I}^{\text{den}}$.

$$\hat{I}^{\text{den}} = \hat{I}^{\text{A}} - \hat{I}^{\text{B}}. \tag{7}$$

Summary Model A provides temporally consistent reconstructions and robust segmentation through stochastic supervision, while Model B performs residual refinement to better align with the challenge objectives. Together, this dual-model design (42.4 M parameters total) combines structural preservation with appearance adaptation.

2.4 Used Loss-Functions

Table 1. Overview of employed loss functions, grouped by type (ROI vs. non-ROI) and level (per-sample vs. batch).

Loss	**Type**	**Level**	**Purpose**
Soft-KS	ROI-based	per-sample	Match / separate ROI histograms
CNR	ROI-based	per-sample	Maximize ROI contrast-to-noise
gCNR	ROI-based	per-sample	Minimize ROI histogram overlap
Alignment	non-ROI	per-sample	Structural similarity (MAE, MS-SSIM, cos)
Center-Triplet	non-ROI	per-sample	Pull to clean, push away from noisy center
Patch Mean/Var	non-ROI	per-sample	Match local patch mean/variance
FID (Mini-FID)	non-ROI	batch	Align batch mean/cov to clean stats
Pixel Hist	non-ROI	batch	Match global pixel histogram

Soft-KS Loss Pixel intensities are softly binned with Gaussian kernels (K=64, σ=0.02) to obtain differentiable histogram $h(x)$, from which cumulative distributions $c(x)_k$ are computed. For two pixel sets x, y we define the soft KS distance:

$$d(x, y) = \| c(x) - c(y) \|_2. \tag{8}$$

Given an ROI mask M with regions A (ventricular septum) and B (ventricle), the final loss is:

$$\mathcal{L}_{\text{KS}}(I^{\text{noisy}}, \hat{I}^{\text{den}}, M^{\text{noisy}}) = d(I_A^{\text{noisy}}, \hat{I}_A^{\text{den}}) - d(I_B^{\text{noisy}}, \hat{I}_B^{\text{den}}) \tag{9}$$

CNR Loss For two disjoint regions with means μ_A, μ_B and standard deviations σ_A, σ_B we calculate the contrast-to-noise ratio loss as:

$$\mathcal{L}_{\text{CNR}}(I, M) = -\frac{|\mu_A - \mu_B|}{\sqrt{\sigma_A^2 + \sigma_B^2 + \varepsilon}} \tag{10}$$

gCNR Loss Let h_A, h_B denote the differentiable Gaussian–soft histograms (K=32, σ=0.02) of the two regions. The generalized CNR loss is:

$$\mathcal{L}_{\mathrm{gCNR}}(I, M) = -\left(1 - \sum_{k=1}^{K} \min(h_{A,k}, h_{B,k})\right), \tag{11}$$

with I the image and M the ROI mask.

Alignment Loss We combine mean absolute error (MAE), multi-scale structural similarity (MS-SSIM), and a cosine-similarity term

$$L_{\cos}(x, y) \;=\; 1 - \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \, \|\mathbf{y}\|_2 + \varepsilon}, \tag{12}$$

where $\mathbf{x}, \mathbf{y}$ are vectorized images. The final objective is a weighted sum:

$$\mathcal{L}_{\mathrm{align}}(x, y) = \lambda_1 \, \mathrm{MAE}(x, y) + \lambda_2 \left(1 - \text{MS-SSIM}(x, y)\right) + \lambda_3 \, L_{\cos}(x, y) \tag{13}$$

The alignment loss is designed to promote consistency between the input and output images. We apply this loss in two settings: (i) $\mathcal{L}_{\mathrm{align}}(\hat{I}^{\mathrm{den}}, I^{\mathrm{noisy}})$ and (ii) $\mathcal{L}_{\mathrm{align}}(\hat{I}^{\mathrm{den}}, \hat{I}^{\mathrm{pca}})$, with default weights $\lambda_1 = 1.0, \lambda_2 = 0.2, \lambda_3 = 0.2$.

Center–Triplet Loss [2] Let $f(\cdot)$ be a frozen Inception-v3 backbone [15] (up to *pool3*, D=2048) that embeds an image I as $z = f(I) \in \mathbb{R}^D$. Using separate sets of I^{clean} and I^{noisy} with standard ImageNet preprocessing we precompute class statistics:

$$\mu_c = \frac{1}{N_c} \sum_{i=1}^{N_c} z_i, \quad c \in \{\text{clean}, \text{noisy}\} \tag{14}$$

$$\Sigma_c = \frac{1}{N_c} \sum_{i=1}^{N_c} z_i z_i^\top - \mu_c \mu_c^\top, \quad \sigma_c^2 = \mathrm{diag}(\Sigma_c). \tag{15}$$

During training, with p=2, the distances to the clean and noisy centers are:

$$d_{\mathrm{pos}}(z) = \|z - \mu_{\mathrm{clean}}\|_2, \qquad d_{\mathrm{neg}}(z) = \|z - \mu_{\mathrm{noisy}}\|_2 \tag{16}$$

The margin-based objective is:

$$\mathcal{L}_{\mathrm{ctrip}}(\hat{I}^{\mathrm{den}}) = \max\!\big(0,\, d_{\mathrm{pos}}(f(\hat{I}^{\mathrm{den}})) - d_{\mathrm{neg}}(f(\hat{I}^{\mathrm{den}})) + m\big), \quad m = 1.0 \tag{17}$$

Additionally (after a warmup phase), we regularize the batch variance of embeddings towards the clean variance vector:

$$\mathcal{L}_{\mathrm{var}}(\hat{I}^{\mathrm{den}}) \;=\; \Big\| \mathrm{diag}\big(\hat{\Sigma}[f(\hat{I}^{\mathrm{den}})]\big) - \sigma^2_{\mathrm{clean}} \Big\|_1, \tag{18}$$

The final loss is:

$$\mathcal{L}_{\mathrm{trip}}(\hat{I}^{\mathrm{den}}) = \mathcal{L}_{\mathrm{ctrip}}(\hat{I}^{\mathrm{den}}) + w_e \, \mathcal{L}_{\mathrm{var}}(\hat{I}^{\mathrm{den}}), \quad w_e = 0.1 \tag{19}$$

The center triplet loss should provide a consistent per-sample pull towards the clean manifold and away from the noisy one.

FID Loss Like in the center–triplet loss, we use Inception-v3 pool-3 features with precomputed clean statistics $(\mu_{\text{clean}}, \Sigma_{\text{clean}})$. In contrast to the sample-wise triplet objective, the FID loss enforces that the *entire batch distribution* of denoised features matches the clean distribution:

$$\mathcal{L}_{\text{FID}}(\hat{I}^{\text{den}}) = \frac{1}{D} \parallel \mu_d - \mu_{\text{clean}} \parallel_2^2 + \frac{1}{D} \parallel \Sigma_d - \Sigma_{\text{clean}} \parallel_F, \quad D = 2048, \tag{20}$$

with (μ_d, Σ_d) denoting the empirical batch mean and covariance. Inception-v3 pool-3 features serve as a low-cost surrogate for the standard FID feature space.

Patch Loss Each image is partitioned into a 16×16 grid, and per-cell mean and variance are precomputed from the clean training set. On a *per-image* basis, the loss $\mathcal{L}_{\text{patch}}$ penalizes squared deviations of denoised cell statistics from these references, thereby enforcing local texture consistency.

Pixel Histogram Loss We precompute the global clean pixel histogram (256 bins, intensities $< 7/255$ ignored). On a *batch level*, the loss $\mathcal{L}_{\text{hist}}$ measures the Z-score–normalized deviation between the denoised and clean histograms, thereby enforcing consistency of the global intensity distribution.

2.5 Training Strategy

The training of the overall system was divided into multiple phases, each designed to address specific challenges and progressively improve the network performance. The implementation is based on *PyTorch* [6], with Inception models provided by *torchvision* [16] and the UNet++ architecture taken from *Segmentation Models Pytorch* [12]. For optimization, we use the AdamW [9] optimizer and a batchsize of 64 together with a cosine annealing learning rate schedule with warm restarts [10].

Phase 1: Initial Training of Model A
In the first stage, Model A was trained with the objective of jointly predicting the PCA-based reconstruction $\hat{I}^{\text{pca}}$ and the binary segmentation $\hat{I}^{\text{seg}}$. For this purpose, the previously described data generator, which integrates PCA-based noise and target augmentation through varying thresholds during training, was employed.

A curriculum learning strategy was applied with respect to the noise magnitude γ: training was initiated with low scaling factor $\gamma = 0.05$ $(m_j = 10, \sigma = 0.75)$ that were gradually increased up to $\gamma = 0.75$ over a total of 500 epochs. In addition, conventional data augmentation techniques (Table 2) were applied. A dedicated augmentation step was introduced, in which the bottom 0-25% of the image was masked (set to 0) with an initially high probability (90%). This probability was progressively reduced during training until a final value of 10% was reached.

Table 2. Overview of employed data augmentation techniques and probabilities during different training phases.

Augmentation	Parameters	Probability	Phase
Resize (scaling)	Scale factor $\in$ [0.93, 1.07]	$p = 0.2$	1,2
Shift (translation)	Max. displacement: $(\pm 25, \pm 25)$ px	$p = 0.1$	1,2,3
Cutout (random erasing)	$n \in [2, 6]$ holes, size $\in [16, 48]$ px	$p = 0.1$	1,2,3
Gaussian blur	Kernel size = 5, $\sigma = 1.5$	$p = 0.1$	1,2,3
Horizontal flip	Mirror along vertical axis	$p = 0.1$	1,2,3
Contrast adjustment	Factor $\in$ [0.9, 1.1]	$p = 0.1$	1,2,3
Mean shift (intensity)	Shift $\in$ [−10, 10]	$p = 0.1$	1,2,3
Bottom masking	Fraction hidden $\in [0.00, 0.25]$	$p = 0.9$ to 0.1	1,2,3
Resize (scaling)	Scale factor $\in$ [0.98, 1.02]	$p = 0.05$	4,5
Shift (translation)	Max. displacement: $(\pm 2, \pm 2)$ px	$p = 0.05$	4,5
Cutout (random erasing)	$n \in [2, 6]$ holes, size $\in [16, 48]$ px	$p = 0.05$	4,5
Gaussian blur	Kernel size = 5, $\sigma = 1.5$	$p = 0.05$	4,5
Horizontal flip	Mirror along vertical axis	$p = 0.05$	4,5
Contrast adjustment	Factor $\in$ [0.99, 1.01]	$p = 0.05$	4,5
Mean shift (intensity)	Shift $\in$ [−2, 2]	$p = 0.05$	4,5
Bottom masking	Fraction hidden $\in [0.00, 0.25]$	$p = 0.01$	4,5

Both components of the curriculum learning procedure (progressive increase in noise amplitude, masking of lower image regions) proved to be essential. Without them, the network quickly starts overfitting to the ECG traces (frequently present in the bottom image area), thereby neglecting the cardiac structures of interest (kind of cheating).

The per sample training loss consist of MSE (for the pca regression) and BCE (for binary segmentation). To account for differences in annotation quality, training samples were weighted w_i individually over all training phases with:

- $w_i = 15$: frames with ground-truth masks
- $w_i = 3$: frames with interpolated masks
- $w_i = 1$: frames without masks (initial clean images)

Phase 2: Auxiliary Network Training
Following Phase 1, an auxiliary step was introduced to generate target regions of interest (ROIs) for the "clean" images also. For this purpose, an independent model H was trained, initialized via transfer learning from model A. The auxiliary network was adapted to treat regions A (ventricular septum) and B (ventricle) as independent segmentation problems, with the goal of maximizing the intersection-over-union (IoU) between denoised input images ($\hat{I}^{\mathrm{A}}$) and their corresponding ground-truth or interpolated masks. After completion of this training step, interpolated masks were generated for the clean images, enabling their integration into the subsequent training stages.

Phase 3: Extended Training of Model A
In this phase, Model A was trained for an additional 250 epochs. During this

stage, the probability of masking the bottom image regions was systematically reduced, encouraging the network to focus on structural rather than contextual cues. The output of Model A after this phase was characterized by high-contrast reconstructions with notably smooth structural regions.

Phase 4: Training of Model B
The second major stage involved the training of Model B, which aimed to refine the outputs of Model A by aligning them more closely with the challenge objectives. Appearance of the final denoised output should be similar to the clean images (FID) while simultaneously improving quantitative metrics such as CNR, gCNR and KS (Kolmogorov–Smirnov).

During this phase, Model A was frozen, and Model B was trained for 250 epochs. Model B was designed to predict a residual (delta) image, which was subtracted from the output of Model A. The relative weighting of the different loss components was adjusted iteratively based on both quantitative loss values and qualitative inspection of the denoised images.

The overall training objective is a weighted sum of all partial losses, each controlled by coefficients $\phi_1, \ldots, \phi_8$:

$$\mathcal{L}_{\text{total}} = \begin{array}{l} \phi_1 \, \mathcal{L}_{\text{KS}} + \phi_2 \, \mathcal{L}_{\text{CNR}} + \phi_3 \, \mathcal{L}_{\text{gCNR}} + \\ \phi_4 \, \mathcal{L}_{\text{align-noisy}} + \phi_5 \, \mathcal{L}_{\text{align-pca}} + \phi_6 \, \mathcal{L}_{\text{trip}} + \\ \phi_7 \, \mathcal{L}_{\text{FID}} + \phi_8 \, \mathcal{L}_{\text{grid}} + \phi_9 \, \mathcal{L}_{\text{phist}} \end{array} \tag{21}$$

Phase 5: Joint Fine-tuning of Models A and B
In the final stage, Models A and B were jointly fine-tuned. The loss function parameters were adapted based on early feedback from the challenge evaluation, with the final configuration set to $\phi_1 = 0.09$, $\phi_2 = 0.07$, $\phi_3 = 0.05$, $\phi_4 = 2.5$, $\phi_5 = 2.5$, $\phi_6 = 2$, $\phi_7 = 50$, $\phi_8 = 1.0$, $\phi_9 = 0.5$. Furthermore, the input noise amplitude was reduced during this stage, as high amplitudes were found to hinder the fine-tuning of the Fréchet Inception Distance (FID) score.

3 Results

Figure 3 illustrates qualitative examples of our algorithm. The denoised images show reduced noise levels and enhanced contrast between the ventricular region and surrounding tissue, while the main cardiac structures remain preserved without introducing structural artifacts. The ROI overlays highlight relevant subregions and demonstrate that interpolated masks align reasonably well with the underlying anatomy, suggesting that the interpolation strategy provides sufficiently accurate labels for subsequent analysis.

Table 3 summarizes the final quantitative results on the test set. The FID score of 58.16 indicates that the proposed algorithm maintains typical echocardiographic structures, as lower values correspond to greater similarity to the clean reference distribution. Both CNR and gCNR improve compared to the noisy inputs, reflecting better separability between septum and ventricle. The KS-based metric shows a low distance (KS-A) between noisy and denoised septal regions, aligning with the challenge objectives.

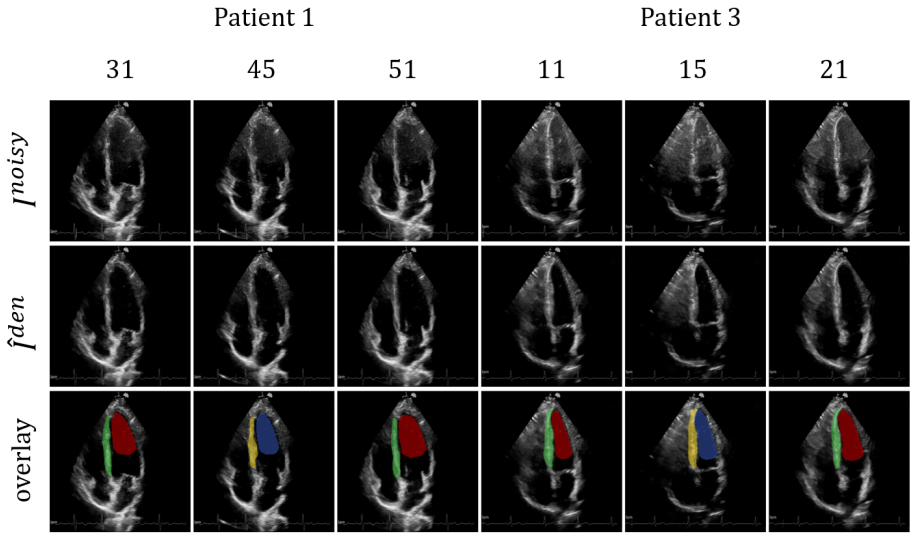

Fig. 3. Noisy (I^{noisy}) and denoised ($\hat{I}^{\text{den}}$) images for two patients across different frame indices (first and second row). The third row shows the denoised images with ROI overlays. Where green/red denote the original masks and the yellow/blue the interpolated masks.

Table 3. Challenge results of the proposed algorithm

FID	CNR	gCNR	ASD	Dice	KS-A	KS-B	Final score
58.1563	2.0351	0.8491	0.21	0.8928	0.0518	0.6646	**84.28**

4 Discussion

Our results indicate that the proposed algorithm reduces noise while preserving the main anatomical structures of the heart. Qualitative examples further suggest that ROI interpolation provides sufficiently accurate labels for downstream analysis.

The weighting of the loss terms allows flexible adjustment of the output towards different evaluation metrics (FID, CNR, gCNR, KS-A, KS-B). However, some objectives are partly antagonistic: for instance, improving CNR often comes at the cost of a higher KS-A score. In our experiments, we prioritized FID, CNR, and gCNR, as these contributed most strongly to the official challenge score. In hindsight, we were not aware that the individual metrics had hard saturation limits, beyond which further improvements did not increase the final score. This strict weighting scheme therefore introduced a misalignment between our optimization focus and the actual evaluation criteria.

A further observation is that the ventricular region in our reconstructions may appear overly smoothed compared to the provided clean references. Nev-

ertheless, an additional advantage of our framework is its ability to denoise not only noisy but also clean images, thereby expanding its potential utility.

Future work will focus on refining the loss weighting strategy to better account for the saturation limits of the challenge metrics and on optimizing ROI predictions to further improve the Dice score of the provided evaluation model. In addition, further work is required to reduce errors captured by the ASD metric. Moreover, the current overload of loss terms allows addressing individual factors explicitly, but a systematic evaluation is needed to identify the truly decisive loss components.

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